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Part manipulator for assembly machine

Abstract

A part manipulator an end effector coupled to a distal end of a robot arm. The end effector includes first and second clamp assemblies coupled to a frame. The first clamp assembly includes a first part gripper configured to pick a first part. The second clamp assembly includes a second part gripper configured to pick a second part. The second part gripper is movable relative to the frame in an actuation direction between a retracted position and an extended position. The second part gripper is located a second distance from the distal end of the robot arm in the retracted position. The second part gripper is located a third distance from the distal end of the robot arm in the extended position greater than the second distance.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application claims benefit to Chinese Application No. 202310121284.9, filed 15 Feb. 2023, the subject matter of which is herein incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

- (2) The subject matter herein relates generally to part assembly machines.
- (3) Part assembly machines are used to assemble parts into products using machine building processes rather than manual, hand building processes. Part assembly machines reduce assembly time and cost. However, automated assembly are not without disadvantages. For instance, pick and place machines have cycle times in moving the parts between the pick location and the place location. Time is wasted in orienting the gripper head and in transitioning between the pick location and the place location. Such transitions take time and reduce throughput of the part assembly machine.
- (4) A need remains for a part assembly machine that may be operated in a cost effective and reliable manner.

BRIEF DESCRIPTION OF THE INVENTION

(5) In one embodiment, a part manipulator is provided and includes a robot arm movable in three dimensional space. The robot arm is movable between a pick station and a place station. The part manipulator includes an end effector coupled to a distal end of the robot arm. The end effector

includes a frame, a first clamp assembly coupled to the frame, and a second clamp assembly coupled to the frame. The first clamp assembly includes a first part gripper configured to pick a first part. The first part gripper is located a first distance from the distal end of the robot arm to pick the first part. The second clamp assembly includes a second part gripper configured to pick a second part. The second part gripper is movable relative to the frame in an actuation direction between a retracted position and an extended position. The second part gripper is located a second distance from the distal end of the robot arm in the retracted position. The second distance is less than the first distance. The second part gripper is located a third distance from the distal end of the robot arm in the extended position. The third distance is greater than the second distance. The second part gripper is configured to pick the second part when the second part gripper is in the extended position.

(6) In another embodiment, a part assembly machine is provided and includes a pick station having a part feeder. The part feeder has a platform supporting parts include a first part and a second part. The part assembly machine includes a vision inspection station positioned adjacent the part feeder. The vision inspection station includes an imaging device to image the first part and the second part in a field of view above the platform. The part assembly machine includes a controller receiving images from the imaging device. The controller determines locations of the first part and the second part on the platform. The part assembly machine includes a part manipulator positioned adjacent the pick station to successively pick up the first part and the second part from the part feeder and configured to successively place the first part and the second part at a place station. The part manipulator includes a robot arm and an end effector is coupled to a distal end of the robot arm. The robot arm operably coupled to the controller. The robot arm is movable in three dimensional space between the pick station and the place station. The end effector operably coupled to the controller. The end effector includes a frame, a first clamp assembly coupled to the frame, and a second clamp assembly coupled to the frame. The first clamp assembly includes a first part gripper configured to pick the first part. The first part gripper is located a first distance from the distal end of the robot arm to pick the first part. The second clamp assembly includes a second part gripper configured to pick the second part. The second part gripper is movable relative to the frame in an actuation direction between a retracted position and an extended position. The second part gripper is located a second distance from the distal end of the robot arm in the retracted position. The second distance is less than the first distance. The second part gripper is located a third distance from the distal end of the robot arm in the extended position. The third distance is greater than the second distance. The second part gripper is configured to pick the second part when the second part gripper is in the extended position. The controller operates the robot arm to successively position the end effector relative to the first part and the second part. The controller operates the first clamp assembly to pick up the first part at the pick station when the first part gripper is aligned with the first part. The controller operates the second clamp assembly to move the second part gripper to the extended position to pick up the second part at the pick station when the second part gripper is aligned with the second part. The controller operates the robot arm to move the end effector to the place station after the first part and the second part are picked up. The controller operates the end effector to successively release the first part and the second part at the place station.

(7) In a further embodiment, a method of assembling parts is provided and includes loading parts, including a first part and a second part, on an upper surface of a platform of a part feeder. The method images the first part and the second part on the platform using an imaging device. The method processes images to determine positions of the first part and the second part on the platform. The method successively picks up the first part and the second part using a part manipulator includes a robot arm is movable in three dimensional space and an end effector is coupled to a distal end of the robot arm that includes a frame, a first clamp assembly coupled to the frame to pick the first part, and a second clamp assembly coupled to the frame to pick the second

part. The first clamp assembly includes a first part gripper is located a first distance from the distal end of the robot arm. The second clamp assembly includes a second part gripper movable relative to the frame in an actuation direction between a retracted position and an extended position. The second part gripper is located a second distance less than the first distance from the distal end of the robot arm in the retracted position. The second part gripper is located a third distance greater than the second distance from the distal end of the robot arm in the extended position. The method operates the robot arm to move the end effector and the first and second parts to a place station and operates the end effector to release the first and second parts at the place station.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic illustration of a part assembly machine for assembling parts, such as parts used to form electrical connectors in accordance with an exemplary embodiment.
- (2) FIG. 2 is a top view of the part assembly machine in accordance with an exemplary embodiment.
- (3) FIG. 3 is a front perspective view of a portion of the part manipulator in accordance with an exemplary embodiment.
- (4) FIG. 4 is a front view of a portion of the part manipulator in accordance with an exemplary embodiment.
- (5) FIG. 5 is a side view of a portion of the part manipulator in accordance with an exemplary embodiment.
- (6) FIG. 6 is a rear view of a portion of the part manipulator in accordance with an exemplary embodiment showing the second part gripper in a retracted position.
- (7) FIG. 7 is a rear view of a portion of the part manipulator in accordance with an exemplary embodiment showing the second part gripper in an extended position.
- (8) FIG. 8 is a front view of a portion of the part manipulator in accordance with an exemplary embodiment showing the second part gripper in the extended position.
- (9) FIG. 9 is a front view of a portion of the part manipulator **200** in accordance with an exemplary embodiment showing the second part gripper **430** in the retracted position.
- (10) FIG. **10** is a front view of a portion of the part manipulator in accordance with an exemplary embodiment.
- (11) FIG. **11** is a flow chart showing a method of assembling parts in accordance with an exemplary embodiment.

DETAILED DESCRIPTION OF THE INVENTION

(12) FIG. 1 is a schematic illustration of a part assembly machine **100** for assembling parts **10**, such as parts used to form electrical connectors. For example, the parts **10** may be contacts, housings, gaskets, seals, wires, circuit boards, or other types of parts. The part assembly machine **100** may be used for assembling products used in other industries. The part assembly machine **100** includes one or more forming machines **30** at a forming station **32** used to form various parts **10**. For example, the forming machines **30** may include a molding machine, a press, a lathe, and the like. The part assembly machine **100** includes one or more processing machines **40** at a processing station **42** used for processing the various parts **10**. For example, the processing station **42** may include an assembly station, a part loading station, a part soldering station, a part termination station, a part packaging station, and the like. The processing machine defines a place station **44** for placing the parts **10**, such as in another product, on another product, or in a package.

(13) The part assembly machine **100** includes a part feeder **102** that supports the parts **10**, such as for transport and/or inspection between the forming machine **30** and the processing machine **40**. The part feeder **102** is used to feed or move the parts **10** through the part assembly machine **100**. In

an exemplary embodiment, the parts **10** may be loaded onto the part feeder **102**. The part assembly machine **100** is able to support the parts **10** without the need for fixturing, which increases the throughput of the parts **10** through the part assembly machine **100**. The parts **10** are picked up from the part feeder **102**. As such, the part feeder **102** defines a pick station **50**, at which the parts **10** are picked.

(14) In an exemplary embodiment, the part assembly machine **100** includes a vision inspection station **110** having one or more imaging devices **112** that image the parts **10** on the part feeder **102** within a field of view of the imaging device(s) **112**. In the illustrated embodiment, the vision inspection station **110** includes multiple imaging devices **112** for imaging different sides of the parts **10**. The imaging device **112** is able to image the parts **10** in the random orientations. In an exemplary embodiment, the vision inspection station **110** may be used to inspect different types of parts **10**. For example, the vision inspection station **110** may be used to inspect different sized parts, different shaped parts, parts in different orientations, and the like.

(15) In an exemplary embodiment, the part assembly machine **100** includes a controller(s) **120** for controlling operation of the various components of the part assembly machine **100**. The controller **120** receives the images from the imaging device **112** and processes the images to determine inspection results. For example, the controller **120** determines the positions and/or orientations of each of the parts **10** on the parts feeder **102**. The controller **120** may inspect the parts, such as for quality and may reject parts that are defective. In an exemplary embodiment, the controller **120** includes a shape recognition tool configured to determine the positions and/or orientations of the parts **10** in the field of view on the parts feeder **102**. The images may be processed by performing pattern recognition of the images based on an image analysis model. The shape recognition tool may compare shapes, patterns or features in the images to shapes, patterns or features in the image analysis model. The images may be processed by performing feature extraction of boundaries and surfaces detected in the images and comparing the boundaries and surfaces to the image analysis model. The controller **120** may identify lines, edges, bridges, grooves, or other boundaries or surfaces within the image. The processing of the images may provide image contrast enhancement for improved boundary or surface identification. In an exemplary embodiment, the controller **120** includes an artificial intelligence (AI) learning module used to customize and configure image analysis based on the images received from the imaging device **112**. The controller **120** may be updated and trained in real time during operation of part assembly machine **100**. For example, the AI learning module may update and train the controller **120** in real time during operation of the vision inspection station **110**.

(16) The vision inspection station **110** includes a part manipulator **200** for moving the parts **10**, such as from the parts feeder **102** to the processing machine **40**, based on the inspection results. For example, the part manipulator **200** may pick up the parts **10** from the parts feeder **102** and place the parts **10** at the processing machine **40**, such as for assembly. In an exemplary embodiment, the part manipulator **200** may be a multi-axis robot manipulator configured to grip and pick the parts off of the parts feeder **102** and move the parts **10** in three-dimensional space. In an exemplary embodiment, the part manipulator **200** is configured to simultaneously move multiple parts **10** (for example, a first part and a second part) from the pick station **50** to the place station **44**. The part manipulator **200** improves efficiency by picking multiple parts **10** and simultaneously transitioning multiple parts between the pick station **50** and the place station **44** by reducing the number of transitions into/out of the parts feeder **102** and/or the processing machine **40**. In various embodiments, the part manipulator **200** may improve cycle time up to 80% compared to part manipulators that move a single part.

(17) FIG. 2 is a top view of the part assembly machine **100** in accordance with an exemplary embodiment. The part assembly machine **100** includes the parts feeder **102**, the vision inspection station **110**, the controller **120**, and the part manipulator **200**.

(18) In an exemplary embodiment, the parts feeder **102** includes a platform **104** and a part feeding

device **106**. The parts **10** are loaded onto the platform **104** by the part feeding device **106**, which may include a hopper, a conveyor, a robot, or another type of feeding device. The parts **10** are presented to the inspection station **110** on the platform **104**. The parts **10** may be advanced or fed along the platform **104**, such as by vibration of the platform **104**. The parts **10** are removed from the platform **104** by the part manipulator **200**. The platform **104** may include a plate having an upper surface **108** used to support the parts **10**. The platform **104** may be a vibration tray that is vibrated to advance the parts **10**. The platform **104** may be rectangular. However, the platform **104** may have other shapes in alternative embodiments, such as a round shape. The platform **104** may rotate.

(19) The inspection station **110** includes one or more imaging devices **112** (a single imaging device **112** is illustrated in FIG. 2) arranged adjacent the platform **104**. The imaging device **112** may be located above the upper surface **108** and view the parts **10** arranged on the upper surface **108**. The imaging device **112** may be a camera, such as a visible light camera, an infrared camera, and the like. The field of view of the imaging device **112** may include the entire surface of the platform **104**. The field of view may include at least a first part **10a** and a second part **10b**. The imaging device **112** may be mounted to a position manipulator **114** for moving the imaging device **112** relative to the platform **104**. The position manipulator **114** may be an arm or a bracket that supports the imaging device **112**. In various embodiments, the position manipulator **114** may be positionable in multiple directions, such as in two-dimensional or three-dimensional space. The position manipulator **114** may be automatically adjusted, such as by the controller **120**, to control positioning of the position manipulator **114**. In other various embodiments, the position manipulator **114** may be manually adjusted. The position of the imaging device **112** may be adjusted based on the types of parts **10** being imaged. For example, when a different type of part **10** is being imaged, the imaging device **112** may be moved based on the type of part being imaged.

(20) The part manipulator **200** is positioned adjacent the platform **104**. The part manipulator **200** is used to pick up the parts **10**, such as the first part **10a** and the second part **10b**, and move the parts **10a**, **10b** to the place station **44** (shown in FIG. 1). The part manipulator **200** picks up the parts **10** based on input from the imaging device **112**. In an exemplary embodiment, the part manipulator **200** includes a robot arm **210** and an end effector **220** at a distal end **212** of the robot arm **210**. The end effector **220** may be a mechanical gripper or vacuum gripper configured to pick up the part **10**. In various embodiments, the robot arm **210** is a four-axis robot arm or a six-axis robot arm. Other types of robot arms may be used in alternative embodiments. The parts **10** are picked up off of the platform **104** by the end effector **220**. The parts feeder **102** may be operated to change the positions and/or orientations of the parts **10**, such as vibrating the platform **104** to change the positions and/or orientations of the parts **10**. The part manipulator **200** may then be operated again to pick up the newly positioned parts **10**.

(21) The controller **120** includes one or more processors **122** for processing the images. The controller **120** is operably coupled to the imaging device **112** and the part manipulator **200** for controlling operation of the part manipulator **200**. The imaging device **112** communicates with the controller **120** through machine vision software to process the data, analyze results, record findings, and make decisions based on the information. The controller **120** provides consistent and efficient inspection automation. The controller **120** determines the positions of the parts **10** to determine which parts **10** are ready to be picked and placed by the part manipulator **200**. The controller **120** controls operation of the part manipulator **200** based on the identified locations (x, y, z) and orientations (for example, heading and facing directions) of the parts **10**. The controller **120** includes a communication module **124** for communicating with the various components of the part assembly machine **100**. The communication module **124** may communicate via wired connections or wireless communication. In an exemplary embodiment, the controller **120** includes a user interface **126**. The user interface **126** includes a display, such as a monitor. The user interface **126** includes one or more inputs, such as a keyboard, a mouse, buttons, and the like. An operator is able

to interact with the controller **120** with the user interface **126**.

(22) FIG. **3** is a front perspective view of a portion of the part manipulator **200** in accordance with an exemplary embodiment. FIG. **4** is a front view of a portion of the part manipulator **200** in accordance with an exemplary embodiment. FIG. **5** is a side view of a portion of the part manipulator **200** in accordance with an exemplary embodiment. FIGS. **3-5** illustrate the end effector **220** provided at the distal end **212** of the robot arm **210**. FIGS. **4-5** show the end effector **220** holding one of the parts **10**.

(23) In an exemplary embodiment, the end effector **220** includes a mounting bracket **222** and a frame **224** extending from the mounting bracket **222**. The end effector **220** includes a first clamp assembly **300** for picking the first part **10a** and a second clamp assembly **400** for picking the second part **10b**. The end effector **220** may include additional clamp assemblies for picking additional parts **10**. The end effector **220** is moved in three-dimensional space by the robot arm **210**.

(24) The mounting bracket **222** is configured to be mounted to the distal end **212** of the robot arm **210**. In various embodiments, the mounting bracket **222** is manufactured from a metal material, such as steel. The mounting bracket **222** may be machined to include openings, slots, or other features used to support the components of the end effector **220**, such as the frame **224**. The mounting bracket **222** may be secured to the robot arm **210** using bolts, latches, clips, or other mounting features.

(25) The frame **224** includes one or more plates of other structural components used to support the components of the end effector **220**. In various embodiments, the frame **224** is manufactured from a metal material, such as steel. The frame **224** may be machined to include openings, slots, or other features used to support the components of the end effector **220**. The frame **224** may be secured to the mounting bracket **222** using bolts, latches, clips, or other mounting features. In various embodiments, the frame **224** extends generally perpendicular to the mounting bracket **222**. For example, the mounting bracket **222** may be oriented horizontally and the frame **224** extends generally vertically along a frame axis **226**.

(26) The first clamp assembly **300** is coupled to a first side of the frame **224**. The first clamp assembly **300** includes a first clamp mount **310**, a first clamp actuator **320**, and a first part gripper **330**. The first clamp mount **310** is used to mount the first clamp assembly **300** to the frame **224**, such as using fasteners. In various embodiments, the first clamp mount **310** is coupled directly to the frame **224**. The first clamp mount **310** may be fixed relative to the frame **224**. However, in alternative embodiments, an intervening structure may be provided between the first clamp mount **310** and the frame **224**. Such intervening structure may be an actuator that moves the first clamp mount **310** relative to the frame **224**, such as in a vertical direction to allow movement of the first part gripper **330** relative to the frame **224**.

(27) The first clamp actuator **320** is coupled to the first clamp mount **310**. In various embodiments, the first clamp actuator **320** is a pneumatic actuator having flow ports **322**, **324**, which may be connected to pneumatic lines, to allow air flow through the first clamp actuator **320** to control operation of the first clamp actuator **320**. In alternative embodiments, the first clamp actuator **320** is a hydraulic actuator or an electric actuator. The first part gripper **330** is coupled to the first clamp actuator **320**. The first clamp actuator **320** is operated to actuate the first part gripper **330**.

(28) In an exemplary embodiment, the first part gripper **330** is used to mechanically pick up and hold the part **10**, such as for movement of the part **10** from the parts feeder **102** to the assembly station at the processing machine **40**. In the illustrated embodiment, the first part gripper **330** includes a first gripper jaw **332** and a second gripper jaw **334** that may be opened and closed relative to each other. A holding space **336** is defined between the first and second gripper jaws **332**, **334**. The part **10** may be held in the holding space **336** between the first and second gripper jaws **332**, **334**. In an exemplary embodiment, the first clamp actuator **320** is operated to cause the first part gripper **330** to pickup and release the part **10**. The first clamp actuator **320** is operably

coupled to the first gripper jaw **332** and/or the second gripper jaw **334** to open and close the first part gripper **330** for pickup and release of the part **10**. Other types of grippers may be used in alternative embodiments to pick up and hold the part **10**. For example, the first part gripper **330** may include vacuum elements used to hold the part **10** by vacuum pressure.

(29) In an exemplary embodiment, the first clamp actuator **320** is operably coupled to the first part gripper **330** to move the first part gripper **330** between a releasing position and a holding position. For example, the first gripper jaw **332** and/or the second gripper jaw **334** may be moved toward each other as the first part gripper **330** is moved to the holding position. The first part gripper **330** is configured to hold the part **10** in the holding position. The first gripper jaw **332** and/or the second gripper jaw **334** may be moved away from each other as the first part gripper **330** is moved to the releasing position. The first part gripper **330** releases the part **10** when moved to the releasing position.

(30) In an exemplary embodiment, the first clamp actuator **320** includes a gripper sensor **326** used for detecting a position of the first part gripper **330** relative to the part **10**. The gripper sensor **326** may be provided in the holding space **336** between the gripper jaws **332**, **334** to sense positions of the gripper jaw(s) **332**, **334**. The gripper sensor **326** may detect the presence of the part **10** in the holding space **336**. The gripper sensor **326** may be a proximity sensor. Other types of sensors may be used in alternative embodiments, such as pressure sensors.

(31) The second clamp assembly **400** is coupled to a second side of the frame **224**. The second clamp assembly **400** includes a second clamp mount **410**, a second clamp actuator **420**, a second part gripper **430**, and a lateral positioner **450**. The lateral positioner **450** is coupled to the frame **224**. The second clamp mount **410** is used to mount the second clamp actuator **420** and the second part gripper **430** to the lateral positioner **450**, such as using fasteners. The lateral positioner **450** is used to move the second part gripper **430** relative to the frame **224**, such as in a lateral actuation direction **452**. The lateral actuation direction **452** may be a vertical direction. The lateral actuation direction **452** may be in a direction toward/away from the upper surface **108** of the platform **104**.

(32) The second clamp actuator **420** is coupled to the second clamp mount **410**. In various embodiments, the second clamp actuator **420** is a pneumatic actuator having flow ports **422**, **424**, which may be connected to pneumatic lines, to allow air flow through the second clamp actuator **420** to control operation of the second clamp actuator **420**. In alternative embodiments, the second clamp actuator **420** is a hydraulic actuator or an electric actuator. The second part gripper **430** is coupled to the second clamp actuator **420**. The second clamp actuator **420** is operated to actuate the second part gripper **430**.

(33) In an exemplary embodiment, the second part gripper **430** is used to mechanically pick up and hold the part **10**, such as for movement of the part **10** from the parts feeder **102** to the assembly station at the processing machine **40**. In the illustrated embodiment, the second part gripper **430** includes a first gripper jaw **432** and a second gripper jaw **434** that may be opened and closed relative to each other. A holding space **436** is defined between the first and second gripper jaws **432**, **434**. The part **10** may be held in the holding space **436** between the first and second gripper jaws **432**, **434**. In an exemplary embodiment, the second clamp actuator **420** is operated to cause the second part gripper **430** to pickup and release the part **10**. The second clamp actuator **420** is operably coupled to the first gripper jaw **432** and/or the second gripper jaw **434** to open and close the second part gripper **430** for pickup and release of the part **10**. Other types of grippers may be used in alternative embodiments to pick up and hold the part **10**. For example, the second part gripper **430** may include vacuum elements used to hold the part **10** by vacuum pressure.

(34) In an exemplary embodiment, the second clamp actuator **420** is operably coupled to the second part gripper **430** to move the second part gripper **430** between a releasing position and a holding position. For example, the first gripper jaw **432** and/or the second gripper jaw **434** may be moved toward each other as the second part gripper **430** is moved to the holding position. The second part gripper **430** is configured to hold the part **10** in the holding position. The first gripper jaw **432**

and/or the second gripper jaw **434** may be moved away from each other as the second part gripper **430** is moved to the releasing position. The second part gripper **430** releases the part **10** when moved to the releasing position.

(35) In an exemplary embodiment, the second clamp actuator **420** includes a gripper sensor **426** used for detecting a position of the second part gripper **430** relative to the part **10**. The gripper sensor **426** may be provided in the holding space **436** between the gripper jaws **432**, **434** to sense positions of the gripper jaw(s) **432**, **434**. The gripper sensor **426** may detect the presence of the part **10** in the holding space **436**. The gripper sensor **426** may be a proximity sensor. Other types of sensors may be used in alternative embodiments, such as pressure sensors.

(36) The lateral positioner **450** includes a base **454** and a carriage **456** slidably coupled to the base **454**. The carriage **456** is movable in the lateral actuation direction **452**. The base **454** is coupled to the frame **224**. The second clamp mount **410** is coupled to, and movable with, the carriage **456**. The lateral positioner **450** is operated to move the carriage **456**, and thus the second part gripper **430**, in the actuation direction **452** between a retracted position (shown in FIG. 3) and an extended position (shown in FIG. 7). The location of the second part gripper **430** relative to the first part gripper **330** changes as the second part gripper **430** moves between the retracted position and the extended position. In an exemplary embodiment, the second clamp assembly **400** includes a position sensor **458** determining a position of the lateral positioner **450**. For example, the position sensor **458** may determine a position of the carriage **456** relative to the base **454**. The position sensor **458** is used to determine a position of the second part gripper **430**, such as to determine if the second part gripper **430** is in the retracted position or the extended position. In various embodiments, the first clamp assembly **300** may include a lateral positioner similar to the lateral positioner **450** to move the first part gripper **330** relative to the frame **224**.

(37) FIG. 6 is a rear view of a portion of the part manipulator **200** in accordance with an exemplary embodiment showing the second part gripper **430** in a retracted position. FIG. 7 is a rear view of a portion of the part manipulator **200** in accordance with an exemplary embodiment showing the second part gripper **430** in an extended position. The lateral positioner **450** is operated to move the second part gripper **430** between the retracted position and the advanced position. In the illustrated embodiment, the first part gripper **330** is fixed relative to the frame **224**. In alternative embodiments, a lateral positioner may be provided between the first part gripper **330** and the frame **224** to move the first part gripper between a retracted position and an advanced position.

(38) The gripper jaws **332**, **334** of the first part gripper **330** are located a first distance **360** from the distal end **212** of the robot arm **210**. In an exemplary embodiment, in the retracted position (FIG. 6), the gripper jaws **432**, **434** of the second part gripper **430** are located a second distance **460** from the distal end **212** of the robot arm **210**. The second distance **460** is less than the first distance **360**. As such, the first part gripper **330** is able to interface with the parts **10** to pick up the first part **10a** without the second part gripper **430** interfacing with the parts **10**. The second part gripper **430** is in the retracted position to provide clearance and not interfere with the parts **10** as the part manipulator **200** maneuvers the first part gripper **330** into position to pick up the first part **10a**. In an exemplary embodiment, in the extended position (FIG. 7), the gripper jaws **432**, **434** of the second part gripper **430** are located a third distance **462** from the distal end **212** of the robot arm **210**. The third distance **462** is greater than the second distance **460**. In other words, the second part gripper **430** is located further from the distal end **212** of the robot arm **210** in the extended position compared to the retracted position. In an exemplary embodiment, the third distance **462** is greater than the first distance **360**. In other words, the second part gripper **430** is located further from the distal end **212** of the robot arm **210** than the first part gripper **330** in the extended position. As such, the second part gripper **430** is able to interface with the parts **10** to pick up the second part **10b** without the first part gripper **330** interfacing with the parts **10**.

(39) With reference back to FIG. 4, the first part gripper **330** is shown picking the first part **10a**. The second part gripper **430** is shown in the retracted position, at the second distance **460** from the

distal end **212** of the robot arm **210**, in a clearance position so as to not interfere with the parts **10** and allow the first part gripper **330** to pick up the first part **10a**.

(40) FIG. **8** is a front view of a portion of the part manipulator **200** in accordance with an exemplary embodiment showing the second part gripper **430** in the extended position. The second part gripper **430** is shown picking the second part **10b**. The second part gripper **430** is shown in the extended position, at the third distance **462** from the distal end **212** of the robot arm **210** to interface with the parts **10** to pick up the second part **10b** without the first part gripper **330** interfacing with the parts **10**. The first part gripper **330** is in a clearance position when the second part gripper **430** is in the extended position. Both part grippers **330**, **430** hold the corresponding parts **10a**, **10b**. As such, multiple parts **10a**, **10b** are configured to be manipulated and moved by the part manipulator **200**. Additional parts **10** may be moved by utilizing additional part grippers (for example, additional clamp assemblies). The part manipulator **200** improves efficiency by picking multiple parts **10a**, **10b** and simultaneously transitioning the multiple parts **10a**, **10b** by reducing the number of transitions into/out of the parts feeder **102** and/or the processing machine **40**. In various embodiments, the part manipulator **200** may improve cycle time up to 80% compared to part manipulators that move a single part.

(41) FIG. **9** is a front view of a portion of the part manipulator **200** in accordance with an exemplary embodiment showing the second part gripper **430** in the retracted position. The first part gripper **330** is shown holding the first part **10a**. The second part gripper **430** is shown holding the second part **10b**. The part manipulator **200** is operated to move both of the parts **10a**, **10b** to a different location, such as to the place station **44** (shown in FIG. **1**). Both parts **10a**, **10b** may be released at the place station **44**. In other embodiments, the part manipulator **200** may move to the place station **44** with the second part gripper **430** in the extended position (FIG. **8**) rather than the retracted position (FIG. **9**).

(42) FIG. **10** is a front view of a portion of the part manipulator **200** in accordance with an exemplary embodiment. In the illustrated embodiment, both the first clamp assembly **300** and the second clamp assembly **400** include lateral positioners **350**, **450**, respectively. The first part gripper **330** is movable relative to the distal end **212** of the robot arm **210** between a retracted position and an extended position. The first part gripper **330** is shown in an extended position in the illustrated embodiment. The second part gripper **430** is shown in the retracted position. The second part gripper **430** is in a clearance position when the first part gripper **330** is in the extended position to avoid interference when picking the parts **10a**, **10b**. Multiple parts **10a**, **10b** are configured to be manipulated and moved by the part manipulator **200**.

(43) FIG. **11** is a flow chart showing a method of assembling parts in accordance with an exemplary embodiment. The method, at **500**, includes inputting the part numbers or type of parts being assembled by the part assembly machine. The user may manually enter the part numbers or select the type of parts being assembled, such as into a user interface of the part assembly machine. The parts are loaded onto the platform of the parts feeder.

(44) At **502**, the method includes sending control signals to the part manipulator based on the type of parts being assembled. The control signals control positioning of the part manipulator relative to the platform. For example, the control signals control the vertical positioning of the part manipulator (for example, the end effector of the part manipulator) for picking up the parts based on the type of parts being assembled.

(45) At **504**, the controller triggers the imaging device to capture an image of the parts on the platform of the parts feeder. At **506**, the controller processes the images. The image is analyzed by the controller to determine positions and/or orientations of each of the parts. The imaging may be performed quickly and efficiently using the imaging device. The image may be processed using an image analysis model, which is based on the type of parts being assembled. The image analysis model may include a shape recognition tool to determine locations and orientations of the parts. In various embodiments, the images are processed by performing pattern recognition of the images

based on the image analysis model. In various embodiments, the images are processed by performing feature extraction of boundaries and surfaces detected in the images and comparing the boundaries and surfaces to the image analysis model.

(46) At **508**, the controller determines if there are any pickable parts based on the image analysis. The controller determines the number of pickable parts and determines the locations of the pickable parts. At **510**, if there are no pickable parts, the controller sends a signal to the parts feeder to vibrate the parts feeder to flip the parts on the platform and change the positions/orientations of the parts on the platform. After the parts feeder is vibrated, the method returns to step **504** to trigger the camera to capture another image. At **512**, if there are pickable parts, the controller queues the positions (x, y, z) of each of the pickable parts to the parts manipulator and activates motion of the part manipulator.

(47) At **514**, the controller positions the first part gripper above the first part for part pick-up. The controller causes the robot arm to move to a first pick-up staging position. The pick-up staging position may be aligned vertically above the first part. The gripper jaws of the first part gripper are open and aligned with the first part. At **516**, the controller operates the first clamp actuator to close the gripper jaws. The gripper jaws are closed on the first part to hold the first part in the first part gripper.

(48) At **518**, the controller positions the second part gripper above the second part for part pick-up. The controller causes the robot arm to move to a second pick-up staging position. The second pick-up staging position may be aligned vertically above the second part. The gripper jaws of the second part gripper are open. At **520**, the lateral positioner is operated to move the second part gripper from the retracted position to the extended position. The second part gripper is moved downward toward the platform holding the second part. The second part gripper is moved to align the gripper jaws of the second part gripper with the second part. At **522**, the controller operates the second clamp actuator to close the gripper jaws. The gripper jaws are closed on the second part to hold the second part in the second part gripper. At **524**, the lateral positioner is operated to move the second part gripper from the extended position to the advanced position. The second part gripper is moved upward away from the platform.

(49) At **526**, the controller causes the part manipulator to move to a placement position. In an exemplary embodiment, the first part may be positioned at the placement position. The robot arm moves the multiple parts held by the part manipulator to the place station. The controller may cause the robot arm to move to the place station. The controller may cause the robot arm to move to the processing machine, such as to an assembly station. For example, the parts may be assembled with other parts at an assembly machine, such as loading contacts or gaskets into the connector housing. The parts may be attached to another part, such as mounting the parts to a circuit board. At **528**, the first part is released at the place location. The first clamp actuator is operated to open the gripper jaws to release the first part. At **530**, the controller causes the part manipulator to move to a second placement position. The second part may be positioned at the second placement position. At **532**, the lateral positioner is operated to move the second part gripper from the retracted position to the extended position. The second part gripper is moved downward toward the place station to move the second part into position. At **534**, the second part is released at the second place location. The second clamp actuator is operated to open the gripper jaws to release the second part. At **536**, the lateral positioner is operated to move the second part gripper from the extended position back to the retracted position.

(50) At **538**, the controller determines if there are additional pickable parts in the queue. If there are additional parts in the queue, the controller triggers a return to step **514** and positions the first part gripper above the part for part pick-up. If there are no pickable parts in the queue, the controller triggers a return to step **510** vibrate the parts feeder to change the positions/orientations of the parts on the platform. After the parts feeder is vibrated, the method returns to step **504** to trigger the camera to capture another image.

(51) It is to be understood that the above description is intended to be illustrative, and not restrictive. For example, the above-described embodiments (and/or aspects thereof) may be used in combination with each other. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from its scope. Dimensions, types of materials, orientations of the various components, and the number and positions of the various components described herein are intended to define parameters of certain embodiments, and are by no means limiting and are merely exemplary embodiments. Many other embodiments and modifications within the spirit and scope of the claims will be apparent to those of skill in the art upon reviewing the above description. The scope of the invention should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. In the appended claims, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Moreover, in the following claims, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects. Further, the limitations of the following claims are not written in means-plus-function format and are not intended to be interpreted based on 35 U.S.C. § 112(f), unless and until such claim limitations expressly use the phrase “means for” followed by a statement of function void of further structure.

Claims

1. A part manipulator comprising: a robot arm movable in three dimensional space, the robot arm movable between a pick station and a place station; an end effector coupled to a distal end of the robot arm, the end effector including a frame, a first clamp assembly coupled to the frame, and a second clamp assembly coupled to the frame, the first clamp assembly including a first part gripper configured to pick a first part, the first part gripper located a first distance from the distal end of the robot arm to pick the first part, the second clamp assembly including a second part gripper configured to pick a second part, wherein the second part gripper is movable relative to the frame in an actuation direction between a retracted position and an extended position, the second part gripper being located a second distance from the distal end of the robot arm in the retracted position, the second distance being less than the first distance, the second part gripper being located a third distance from the distal end of the robot arm in the extended position, the third distance being greater than the second distance, wherein the second part gripper is configured to pick the second part when the second part gripper is in the extended position.
2. The part manipulator of claim 1, wherein the end effector moves both the first and second parts between the pick station and the place station.
3. The part manipulator of claim 1, wherein the first part gripper is fixed relative to the frame.
4. The part manipulator of claim 1, wherein the second part gripper is movable relative to the first part gripper to pick the second part.
5. The part manipulator of claim 1, wherein the third distance is greater than the first distance.
6. The part manipulator of claim 1, wherein the second part gripper is located above the first part gripper in the retracted position, the second part gripper being located below the first part gripper in the extended position.
7. The part manipulator of claim 1, wherein the first part gripper is movable relative to the frame in an actuation direction between a retracted position and an extended position, the first part gripper being located at the first distance from the distal end of the robot arm in the retracted position, the first part gripper being located a fourth distance from the distal end of the robot arm in the extended position, the fourth distance being greater than the first distance, wherein the first part gripper is configured to pick the first part when the first part gripper is in the extended position.
8. The part manipulator of claim 1, wherein the second clamp assembly includes a lateral positioner coupled to the frame, the lateral positioner being coupled to the second part gripper and operated to

move the second part gripper between the retracted position and the extended position.

9. The part manipulator of claim 1, wherein the second clamp assembly includes a position sensor determining a position of the second part gripper.

10. The part manipulator of claim 1, wherein the first clamp assembly includes a first clamp actuator operably coupled to the first part gripper to move the first part gripper between a releasing position and a holding position, the first part gripper configured to hold the first part in the holding position, the second clamp assembly includes a second clamp actuator operably coupled to the second part gripper to move the second part gripper between a releasing position and a holding position, the second part gripper configured to hold the second part in the holding position.

11. A part assembly machine comprising: a pick station having a part feeder, the part feeder having a platform supporting parts including a first part and a second part; a vision inspection station positioned adjacent the part feeder, the vision inspection station including an imaging device to image the first part and the second part in a field of view above the platform; a controller receiving images from the imaging device, the controller determining locations of the first part and the second part on the platform; and a part manipulator positioned adjacent the pick station to successively pick up the first part and the second part from the part feeder and configured to successively place the first part and the second part at a place station, the part manipulator including a robot arm and an end effector coupled to a distal end of the robot arm, the robot arm operably coupled to the controller, the robot arm movable in three dimensional space between the pick station and the place station, the end effector operably coupled to the controller, the end effector including a frame, a first clamp assembly coupled to the frame, and a second clamp assembly coupled to the frame, the first clamp assembly including a first part gripper configured to pick the first part, the first part gripper located a first distance from the distal end of the robot arm to pick the first part, the second clamp assembly including a second part gripper configured to pick the second part, wherein the second part gripper is movable relative to the frame in an actuation direction between a retracted position and an extended position, the second part gripper being located a second distance from the distal end of the robot arm in the retracted position, the second distance being less than the first distance, the second part gripper being located a third distance from the distal end of the robot arm in the extended position, the third distance being greater than the second distance, wherein the second part gripper is configured to pick the second part when the second part gripper is in the extended position; wherein the controller operates the robot arm to successively position the end effector relative to the first part and the second part; wherein the controller operates the first clamp assembly to pick up the first part at the pick station when the first part gripper is aligned with the first part; wherein the controller operates the second clamp assembly to move the second part gripper to the extended position to pick up the second part at the pick station when the second part gripper is aligned with the second part; wherein the controller operates the robot arm to move the end effector to the place station after the first part and the second part are picked up; and wherein the controller operates the end effector to successively release the first part and the second part at the place station.

12. The part assembly machine of claim 11, wherein the end effector moves both the first and second parts between the pick station and the place station.

13. The part assembly machine of claim 11, wherein the first part gripper is fixed relative to the frame.

14. The part assembly machine of claim 11, wherein the second part gripper is movable relative to the first part gripper to pick the second part.

15. The part assembly machine of claim 11, wherein the second part gripper is located above the first part gripper in the retracted position, the second part gripper being located below the first part gripper in the extended position.

16. The part assembly machine of claim 11, wherein the second clamp assembly includes a lateral positioner coupled to the frame, the lateral positioner being coupled to the second part gripper and

operated to move the second part gripper between the retracted position and the extended position.

17. A method of assembling parts comprising: loading parts, including a first part and a second part, on an upper surface of a platform of a part feeder; imaging the first part and the second part on the platform using an imaging device; processing images to determine positions of the first part and the second part on the platform; successively picking up the first part and the second part using a part manipulator including a robot arm movable in three dimensional space and an end effector coupled to a distal end of the robot arm that includes a frame, a first clamp assembly coupled to the frame to pick the first part, and a second clamp assembly coupled to the frame to pick the second part, the first clamp assembly including a first part gripper located a first distance from the distal end of the robot arm, the second clamp assembly including a second part gripper movable relative to the frame in an actuation direction between a retracted position and an extended position, the second part gripper being located a second distance less than the first distance from the distal end of the robot arm in the retracted position, the second part gripper being located a third distance greater than the second distance from the distal end of the robot arm in the extended position; operating the robot arm to move the end effector and the first and second parts to a place station; and operating the end effector to release the first and second parts at the place station.

18. The method of claim 17, wherein said successively picking up the first part and the second part comprises: positioning the second part gripper in the retracted position when the first part is picked up by the first part gripper; and positioning the second part gripper in the extended position when the second part is picked up by the second part gripper.

19. The method of claim 17, wherein said successively picking up the first part and the second part comprises: moving the robot arm to align the first part gripper with the first part based on the images to pick up the first part; and moving the robot arm to align the second part gripper with the second part based on the images to pick up the second part.

20. The method of claim 17, further comprising: vibrating the parts to change orientations of the parts on the upper surface of the platform; reimaging the parts; and processing new images of the parts to determine new orientations of the parts.
